

Web and Multimedia Technologies

Lab class 2 – DaVinci Resolve

Instructions to use the laboratory computers (room B2/B4):

1. If the computer is off, start it with Windows (all computers have a Linux-Windows dual boot)
2. Your username is *cl* followed by the six digits of your student ID; your password is *inf* followed by the last two digits of your student ID (e.g., if the student ID is 123456, the username is *cl123456* and the password is *inf56*)
3. At the end of the class, please logout

Steps (related to *DaVinci Resolve*, free, *myvideo*):

1. Create the folder *davinci*
2. From <https://vision.unipv.it/wmt/lab/davinci/>, download the files *clip1.mp4*, *clip2.mp4*, *clip3.mp4*, *clip4.mp4*, *clip5.mp4*, and *music.mp3* into the *davinci* folder (in the browser's address bar, write <https://vision.unipv.it/wmt/lab/davinci/clip1.mp4>, etc., and save the video clips and the audio file when they are played – it is not possible to show the list of files in <https://vision.unipv.it/wmt/lab/davinci/>)
3. Open *DaVinci Resolve* and create a new project named “myvideo”
4. With the *Edit* visualization mode selected (in the footer), if the *Media Pool* panel is not visible, click the *Media Pool* button in the upper left corner
5. Drag the five video clips and the audio file from the *davinci* folder to the *Media Pool* area (if a message box is displayed warning that the clips' frame rate is different from the current project's settings, press the *Change* button). A clip can be previewed by double clicking on it
6. Using the mouse, drag *clip1.mp4* to the timeline: now the clip is part of the final video we are building (note that the timeline zoom level can be adjusted through a slider)
7. Drag *clip2.mp4* to the right of *clip1.mp4* in the timeline, attached to it. Move the current-time indicator (vertical red line) to the beginning of *clip1.mp4* and play the video formed of the two clips (button ► in the timeline video preview, on the right)
8. In *clip1.mp4*, we now want to remove the part from time 2:24 to time 4:15: position the time indicator to 2:24 and cut the clip there using the *Blade* tool; then, do the same at 4:15. Using the *Selection* tool, click on the portion of *clip1.mp4* between 2:24 and 4:15 and press DEL/CANC on the keyboard to remove it
9. Click the *Effects* tab (upper left) and select *Video Transitions*
10. From the *Dissolve* group, drag *Cross Dissolve* to the separation between the two sub-clips composing *clip1.mp4*: a cross dissolve effect is introduced as a transition between the sub-clips
11. Now we want to create another transition (e.g., *Heart* from the *Shape* group) between the end of the second part of *clip1.mp4* and the beginning of *clip2.mp4*. However, by positioning the mouse cursor exactly between the two clips and clicking, you can see that the separation is highlighted in red: this means that the transition cannot be applied. You can solve the problem by trimming a little the tail of the first clip and the head of the second clip
12. From the *Media Pool*, drag *clip3.mp4* to the timeline, attached to *clip2.mp4*, and create a transition between them (e.g., *Radial Wipe* from the *Wipe* group). In this case too, trim a little the end of *clip2.mp4* and the beginning of *clip3.mp4* to create the time “handles” for the transition to occur
13. From the *Media Pool*, drag *clip4.mp4* to the timeline, attached to *clip3.mp4*, and, again, create a transition of your choice between them

14. From the *Media Pool*, drag `clip5.mp4` to the timeline and observe its content
15. Click with the right mouse button on `clip5.mp4` and choose *Change Clip Speed...* Change the speed to 300% (the frame rate will become 75 fps, even if, practically, 25 frames will be played in a second – this means that 2/3 of the frames of the original clip will be removed)
16. Click with the right mouse button on the shortened clip `clip5.mp4` and select *Link Clips* to unlink the video and audio tracks of the clip. Then select the audio track and press BACKSPACE on the keyboard to remove it: now we have a mute clip
17. Drag `clip5.mp4` to the video track *V2* above the current video track in the timeline (*V1*), so that the end of `clip5.mp4` is aligned to the end of `clip4.mp4`
18. Select `clip5.mp4` and, in the *Inspector* panel (upper right), change the opacity to 20%: now you can see the content of `clip5.mp4` that is superimposed on the clips that are in track *V2*
19. In the *Effects* library, select the *Titles* section and drag the *Text* title to the beginning of the timeline in track *V2*
20. In the *Inspector* panel (upper right), write “My first video editing” as a title and set its features (e.g., set 120 as a size)
21. Adjust the length of the title as you prefer; by moving the white “handles” in the upper part of the title, adjust the gradual appearance and disappearance of the title
22. In the *Effects* library, select *Resolve FX* and drag a filter to the clip `clip2.mp4` (e.g., *Mosaic Blur*, with 100 as a *Pixel Frequency*): now a “transformation has been applied to the clip
23. From the *Media Pool*, drag `music.mp3` to the audio track *A2*, below the main audio track *A1*, and align it to the left; then cut the audio clip so that it finishes with clips `clip4.mp4` and `clip5.mp4`
24. With the audio clip selected (click on it), open the *Inspector* panel: the volume of the whole clip can be changed (*Clip Volume*)
25. By clicking the “diamond” checkbox at the right of the *Volume* slider in the *Inspector* panel, new audio “nodes” can be created to locally adjust the volume
26. At the end of the clips (both video and audio), create “smooth” endings by dragging the white “handles” (on the right) to the left
27. In the *Deliver* section (lower part), choose *Custom Export* in the *Render Settings* pane (upper left)
28. In the *Video* sub-section, specify filename and location, choose *MP4* as a format, and indicate *H.264* as a codec. As a resolution, choose *1280 x 720 HD 720P* (25 fps, best quality)
29. Press the *Add to Render Queue* button and then, on the right, the *Render All* button, to export the result of your video editing
30. Open the just created video with a video player and check the result